

Stage 2

Computer Vision PROJECT
Umm Al Qura University



Project ID: 3

Project Title: Smart Parking detection with proximity recommendations

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System architecture

Our system is based on intelligent detection of available parking spaces combined with recommending the nearest parking spot. The system is designed as a sequential series of stages to ensure smooth data flow and allow future development and scalability.

The system begins with the parking dataset stage, which contains images or video clips of parking lots captured under different conditions such as variations in lighting and camera angles. After that, the data is loaded into the working environment for processing.

Next, data cleaning is performed. In this stage, corrupted, distorted, duplicated, or low-quality images are removed to prevent negatively affecting the model's performance and to maintain dataset consistency before the training process.

After that, image preprocessing is applied. All images are resized to match the model's input requirements, and pixel values are normalized. After that, techniques such as rotation, brightness adjustment, and flipping are applied to improve the model's generalization and reduce overfitting. Once preprocessing is completed, the dataset is divided into training and validation sets.

Then, the parking space detection stage is performed. In this stage, parking spaces are identified within the image or video using an object detection model to automatically detect parking areas.

After extracting each parking slot individually, the system proceeds to the occupancy classification stage. In this stage, each parking space is analyzed to determine whether it is available or occupied based on the visual information extracted from the image.

Next, the system moves to the decision-making stage. The system first filters only the available parking spaces. Then, it receives the user's location as input and calculates the distance between the user's position and each available parking space. After that, the nearest vacant parking space is selected based on the shortest distance.

After selecting the parking space, the system determines the route to reach it using shortest-path algorithms such as A* search algorithm or Dijkstra's algorithm.

Finally, the results are displayed through the user interface. The parking lot image is shown with occupied and available spaces clearly distinguished, and the driver is guided to the shortest distance with the estimated time of arrival.

so, the architectural design allows future scalability, such as integrating live cameras, which makes the system suitable for real-world applications.

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Algorithms / models to be used

Detecting vacant and occupied parking spaces will involve extracting visual features from images and classifying the occupancy status of each parking spot. Based on this approach, the proposed system in this project will be organized into three main stages: **feature extraction, parking space classification, and nearest parking space determination.**

Stage 1: Feature Extraction

The first stage will focus on extracting meaningful visual features from parking space images. In this project, the **Histogram of Oriented Gradients (HOG)** method will be used for feature extraction. HOG is a widely used computer vision technique that captures the distribution of edge orientations and gradient structures within an image, which will help in identifying objects such as vehicles.

Previous studies have shown that HOG will provide reliable feature representations for vehicle detection when combined with suitable classification algorithms [1].

Some studies have also used deep learning approaches such as **YOLO** for vehicle detection. Although these models can achieve high accuracy, they require greater computational resources and more complex training procedures. In comparison, HOG will provide a computationally efficient solution suitable for this project [2].

Stage 2: Parking Space Classification

After extracting image features, the next stage will involve classifying each parking space as either vacant or occupied. Two machine learning algorithms will be used: **Support Vector Machine (SVM)** and **K-Nearest Neighbors (K-NN)**. SVM is widely used for classification problems involving high-dimensional data and will be applied with HOG features for object detection tasks [1]. K-NN will provide a simple yet effective method for classifying parking spaces based on similarity between feature vectors using distance measurements.

Both classifiers will be trained using the extracted HOG features, and their performance will be compared to determine which algorithm will provide better accuracy in detecting vacant and occupied parking spaces.

Stage 3: Nearest Parking Space Determination

A previous study relied on GPS-based navigation to guide drivers to the nearest available parking space [2]. In contrast, the proposed project will determine the nearest vacant parking spot using an internal representation of the parking layout combined with pathfinding algorithms such as Dijkstra and A*[3].

This stage will focus on computing the shortest path within the parking network itself, enabling efficient identification of the closest available spot. By relying solely on algorithmic pathfinding rather than external positioning data, the approach differentiates the current work from prior research while remaining feasible and computationally efficient for implementation.

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Dataset description

The dataset selected for this project is a pre-processed version of the PKLot dataset available on Kaggle. It is derived from the original PKLot research dataset and adapted to support object detection tasks.

The dataset includes 12,416 images of outdoor parking lots captured by surveillance cameras in three separate locations in Brazil: PUCPR, UFPR04, and UFPR05. The images capture various weather conditions, including sunny, cloudy, and rainy days, providing a robust environment for evaluating computer vision models under different lighting and shadow effects.

Each parking space within an image is manually labeled as "Occupied" or "Empty." The images are in JPEG format, with a resolution of 1280×720. The original annotations have been converted into axis-aligned bounding boxes, which are available in CSV and TXT formats, ensuring that the dataset is fully compatible with standard object detection frameworks.

Milestones and timeline

Week	Milestone	Main Tasks
Week 1	System Setup	Finalize three-stage architecture (HOG detection, SVM/KNN classification, nearest parking logic).
Week 2	Dataset Preparation	Organize PKLot dataset and prepare parking slot samples.
Week 3	Vehicle Detection	Implement HOG for vehicle detection in parking spaces.
Week 4	Occupancy Classification	Train SVM or KNN model to classify spaces as occupied or available.
Week 5	Evaluation	Evaluate model performance and improve accuracy.
Week 6	Nearest Parking Module	Implement distance calculation and Dijkstra algorithm.
Week 7	Integration	Combine detection, classification, and navigation modules.
Week 8	Interface & Testing	Display available spaces, route, distance, and ETA.

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Evaluation metrics

The proposed system will be evaluated based on two main components:

- Parking space detection using a computer vision model.
- Recommendation of the nearest available parking space using path planning algorithms such as A* or Dijkstra.

1. Parking Detection Evaluation

The ability of the model to classify parking spaces as available or occupied will be evaluated using standard classification metrics, including:

- **Accuracy:** Measures the overall percentage of correctly classified parking spaces.
- **Precision:** Indicates how many parking spaces predicted as available are actually available, reducing cases where the system suggests an occupied parking space.
- **Recall:** Measures the ability of the system to correctly detect all available parking spaces.
- **F1-score:** Combines precision and recall providing a balanced evaluation of the model's performance.
- **Confusion Matrix:** Used to analyze classification results and identify different types of prediction errors.

2. Recommendation Accuracy

After detecting the available parking spaces, the system uses A* or Dijkstra algorithms to determine the shortest path from the driver's location to the available parking spots. Based on the computed paths, the system selects the parking space with the lowest path cost and recommends it to the driver.

The recommendation performance will be evaluated using:

- **Recommendation Accuracy:** The percentage of cases in which the system selects the parking space that actually represents the shortest path among all available spaces.
- **Response Time:** The efficiency of the system will also be evaluated by measuring the response time, which represents the total time required for the system to process the input and produce a recommendation.

It includes:

- 1) **Detection Time:** The time required for the model to detect available parking spaces.
- 2) **Path Computation Time:** The time required to calculate the shortest path using A* or Dijkstra.

The average response time will be calculated over multiple test cases.

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Risk assessment

Several potential risks may affect the performance of the proposed system:

- **Lighting variations:** Different lighting conditions such as strong sunlight or shadows may affect parking detection accuracy.
- **Low image quality or noise:** Blurred or noisy images may reduce the model's ability to correctly detect parking spaces.
- **Occlusion:** Vehicles may partially cover parking lines or spaces, making it harder to determine whether a space is available.
- **Dataset imbalance:** If the dataset contains significantly more occupied spaces than available ones, the model may become biased toward one class.
- **Camera angle variations:** Images captured from different angles may change the appearance of parking spaces and affect detection accuracy.
- **Inaccurate parking space representation:** If parking spaces are not represented correctly in the system, A* or Dijkstra may compute incorrect paths and produce inaccurate recommendations.

Fallback plans

To mitigate the impact of these risks, several mitigation strategies will be implemented:

- **Image Quality Improvement:** Techniques such as contrast enhancement and noise reduction can be used to improve image quality before detection [6].
- **Image Preprocessing:** Adjusting brightness and contrast can help reduce the impact of lighting changes.
- **Geometric Transformation:** Processes such as rotation or perspective correction can be used to process images captured from different camera angles [7].
- **Feature or Edge Detection:** These techniques can help highlight the boundaries of parking spaces and vehicle edges [8].
- **Model Adjustment:** Model parameters can be modified, or alternative models can be tested if performance is unsatisfactory.
- **Balanced Dataset:** Maintaining a balanced number of available and occupied parking space samples helps reduce model bias.
- **Improved Parking Layout Representation:** Accurately representing parking spaces helps A* or Dijkstra algorithms determine the shortest path to available parking spots.

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